

TERRACE FARMING MONITORING SYSTEM



22UIT404- A MINI PROJECT I REPORT

Submitted by

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in partial fulfillment for the award of the degree

of

BACHELOR OF TECHNOLOGY

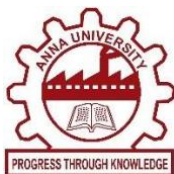
in

INFORMATION TECHNOLOGY

Dr.N.G.P. INSTITUTE OF TECHNOLOGY COIMBATORE- 641 048
(AN AUTONOMOUS INSTITUTION)

ANNAUNIVERSITY: CHENNAI 600 025

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BONAFIDE CERTIFICATE

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INTERNAL EXAMINER

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ABSTRACT

The increasing demand for sustainable agricultural practices necessitates the incorporation of advanced technologies into conventional farming methods. Challenges such as inefficient resource utilization, erratic crop yields and land degradation call for significant changes. The study explores the impact of smart agricultural technologies, including the Internet of Things (IoT), Artificial Intelligence (AI) and autonomous systems, on improving efficiency and sustainability in agriculture. A systematic review of recent studies focused on analyzing existing technologies, methodologies and challenges within this area. Major findings revealed that IoT-based systems primarily enhanced the ability to monitor real-time data, improve predictability and automate processes, leading to better resource utilization, increased crop yields and reduced reliance on labor. Furthermore, AI technologies significantly improved data-driven decision-making and optimized crop management, enabling precision farming techniques, such as optimizing irrigation schedules, selecting the best planting times, and predicting pest outbreaks. Autonomous systems, such as drones and robots, were found to reduce operational costs and labor intensity while enhancing productivity, allowing farmers to focus on higher-level strategic decisions. Smart technologies also contribute to environmental conservation by reducing the overuse of pesticides and fertilizers, minimizing water wastage and reducing carbon footprints. The integration of climate-adaptive tools further supports farmers in dealing with unpredictable weather patterns, enabling more effective responses to climate change. However, challenges remain, including high initial costs, the complexity of technology adoption, limited infrastructure in rural areas and the need for more skilled personnel. Despite these obstacles, integrating IoT and smart technologies offers substantial potential for revolutionizing agriculture, contributing to sustainable practices and enhancing food security.

KEYWORDS: Sustainable agriculture, Smart agricultural technologies, Internet of Things (IoT), Autonomous systems, Crop yield optimization.

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LIST OF ABBREVIATIONS

Abbreviation	Full Form
IoT	Internet of Things
AI	Artificial Intelligence
LCD	Liquid Crystal Display
Blynk	(Brand/Name of IoT Platform/App)
LED	Light Emitting Diode
Node MCU	Node MicroController Unit
Wi-Fi	Wireless Fidelity
UAV	Unmanned Aerial Vehicle
LCD	Liquid Crystall Display
XML	Extensible Markup Language

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CHAPTER 1

INTRODUCTION

1.1 ABOUT THE PROJECT

This project is designed to transform conventional terrace farming by incorporating smart agricultural technologies that support real-time monitoring and precision-based practices. The system uses IoT sensors to track essential environmental factors such as soil moisture, temperature, humidity, and light intensity. This data enables the platform to deliver personalized recommendations for watering, plant care, and general maintenance, making it easier for urban farmers to manage their crops effectively. One of the key goals is to maximize plant growth while reducing the manual effort required, allowing users—regardless of their farming experience—to achieve better outcomes with less guesswork.

A major focus of the project is on sustainable resource management, particularly efficient water usage. By leveraging data-driven insights, the system promotes precision irrigation, helping reduce water waste and improve overall environmental impact. The project also includes the development of a user-friendly mobile application that integrates sensor data with clear, actionable suggestions to streamline the farming process. This approach not only enhances productivity but also encourages urban communities to engage in sustainable agriculture. Ultimately, the project seeks to make terrace farming more accessible, resource-efficient, and environmentally friendly, contributing to food security and the broader adoption of smart farming in urban areas.

1.2 PROBLEM STATEMENT

Terrace farming, while increasingly popular in urban areas, faces several challenges that hinder its efficiency and sustainability. Most conventional terrace farming methods rely heavily on manual observation and generalized practices, which often result in inconsistent plant growth, inefficient use of resources, and increased maintenance efforts. Urban farmers, many of whom are not professionally trained in agriculture, struggle to maintain optimal conditions for their crops due to the lack of real-time monitoring and personalized guidance. This can lead to overwatering, under-maintenance, and resource wastage, particularly in water usage.

Moreover, there is currently a gap in accessible, technology-driven solutions that can support urban farmers in managing their terrace gardens efficiently. Without timely and data-driven insights, users are unable to make informed decisions regarding irrigation, fertilization, and general care. The absence of an integrated, user-friendly platform that combines environmental sensor data with actionable recommendations further complicates the process, making terrace farming less effective and more time-consuming.

Therefore, there is a critical need for a smart farming solution that provides real-time data, delivers personalized recommendations, and simplifies farm management. Such a system would not only improve resource efficiency and crop health but also make terrace farming more accessible and sustainable for urban dwellers

1.3 OBJECTIVE

The main objective of this project is to modernize and improve terrace farming by implementing a smart system that ensures optimal plant growth through real-time environmental monitoring and intelligent decision support. The system will use IoT sensors to collect data on critical parameters such as soil moisture, temperature, humidity, and sunlight. This data will be processed and used to provide personalized, actionable recommendations for tasks like watering, fertilization, and general maintenance. A key goal is to assist urban farmers, particularly those with limited agricultural experience, in making informed decisions to improve crop health and yield. The project also aims to promote sustainability by reducing water wastage and encouraging efficient resource usage through precision farming techniques. To make the technology accessible, a user-friendly mobile application will be developed, combining sensor data with simple, easy-to-understand suggestions. This platform will streamline terrace farm management and reduce the dependency on manual monitoring. Ultimately, the project seeks to empower urban communities to practice sustainable agriculture in limited spaces by making terrace farming more efficient, environmentally friendly, and accessible. By bridging the gap between traditional farming methods and modern technology, the project contributes to food security, resource conservation, and the future of urban farming.

1.4 SCOPE AND LIMITATIONS OF THE STUDY

SCOPE

- Focuses on enhancing terrace farming using smart technologies and IoT-based real-time monitoring.

- Involves the use of sensors to track environmental factors such as soil moisture, temperature, humidity, and light intensity.
- Develops a mobile application to provide users with personalized and actionable recommendations for watering and maintenance.
- Aims to support urban farmers, especially those with limited agricultural knowledge, by simplifying terrace garden management.
- Promotes sustainable farming practices by encouraging efficient use of resources, particularly water.
- Designed primarily for small-scale, domestic terrace farming setups in urban environments.

LIMITATIONS

- Limited to small-scale urban terrace farms; not designed for large-scale agricultural operations.
- Sensor accuracy and performance may be affected by weather conditions and hardware limitations.
- Initial setup costs and technical knowledge may hinder adoption for some users.
- Recommendations may need further adaptation for specific plant types or local environmental conditions.
- Requires stable internet connectivity for real-time updates and full platform functionality.
- Long-term maintenance of hardware (sensors, devices) may present challenges over time.

1.5 ORGANIZATION OF THE REPORT

This report is organized into several chapters to provide a clear and systematic understanding of the project. Chapter 1 introduces the project by presenting the background, problem statement, objectives, scope, and an outline of the report structure. Chapter 2 includes a literature survey that reviews existing research and technologies related to smart terrace farming, highlighting the gaps in current solutions. Chapter 3 discusses the system requirements in detail, covering both hardware and software components essential for implementation. Chapter 4 explains the proposed methodology, system architecture, and the functionalities of each module involved in the monitoring and automation process. Chapter 5 focuses on the system implementation, describing the integration of IoT sensors, microcontroller, and the mobile application to enable real-time monitoring of terrace farming conditions. Finally, Chapter 6 concludes the report by summarizing the key outcomes and offering suggestions for future improvements. The report also includes references and appendices with supporting documents and source code.

CHAPTER 2

LITERATURE SURVEY

2.1 INTRODUCTION ABOUT LITERATURE SURVEY

The agricultural industry is progressively embracing technology to address challenges such as resource inefficiency, environmental sustainability, and food security. Innovations in technology, particularly in IoT, AI and other digital advancements are transforming the landscape of farming. The research will concentrate on how these technologies can be integrated to establish a sustainable and efficient agricultural system. The motivation behind the research arises from resource scarcity and the need for improved crop yields and resource optimization while adhering to sustainable farming practices. Conventional agricultural methods depend heavily on human input, which often leads to inefficiencies, inconsistent yields and environmental harm. With the increasing threats posed by climate change and population growth, IoT, AI and other smart technologies offer solutions for integrating farming practices through real-time monitoring, predictive analysis, and automation, ultimately reducing manual labor and improving decision-making. Additionally, smart technologies can enhance water management, allowing for more precise irrigation and minimizing waste, which is crucial given the growing concern over water scarcity. Precision agriculture, driven by IoT and AI, also allows for targeted fertilization and pest control, further reducing environmental impact and costs. The literature review entailed a methodical examination of peer-reviewed articles and relevant academic research to understand the current landscape of agricultural technologies, their possibilities and challenges.

2.2 REVIEW OF LITERATURE SURVEY

2.2.1. Automatic Greenhouse Monitoring System

Shazeb et al, present the Automatic Greenhouse Monitoring System (AGMS), a solution aimed at improving farming efficiency by tackling key challenges faced in traditional agricultural practices. They describe how the AGMS achieves its objectives by automating the management and regulation of greenhouse conditions, thereby enhancing both crop yield and quality, while also reducing resource consumption. The methodology involves the use of a Node MCU microcontroller, cloud services for processing and storing data, and APIs to integrate various hardware components for real-time data collection and visualization. Key findings suggest that the AGMS significantly enhances monitoring efficiency and user interaction, with a web interface that offers real-time updates. Additionally, there is potential for integrating advanced technologies like machine learning to enable predictive analytics, optimizing greenhouse conditions based on data-driven insights. This study underscores the importance of the AGMS in promoting sustainable farming practices and advancing the application of IoT technologies in agriculture. The findings highlight its relevance within current agricultural research, advocating for the adoption of smart technologies to revolutionize the farming sector.

2.2.2. Sustainable Agriculture in the Digital Age: Crop Management and Yield Forecasting with IoT, Cloud, and AI

Hadeed Ashraf et al, explores the pivotal role of digital technologies in advancing sustainable agricultural practices. The introduction highlights the growing importance of sustainable agriculture and discusses how technologies like the Internet of Things (IoT), Cloud Computing, and Artificial Intelligence (AI) are transforming crop management and yield forecasting. The paper examines how these technologies can address various challenges faced by the agricultural sector while enhancing efficiency and sustainability. In the methodology section, the authors outline the data collection methods and analytical models used to evaluate the impact of these technologies, potentially incorporating case studies to illustrate practical applications. The key findings suggest that the implementation of IoT sensors, cloud computing, and AI-driven analytics has led to increased crop yields, better resource management, and improved decision-making for farmers. The conclusions stress the importance of adopting these technologies to ensure food security and sustainability in the face of climate challenges. Finally, the literature review highlights how the

study contributes to the broader academic discourse on agricultural technology, connecting its findings to existing research and emphasizing the growing relevance of digital tools in modern farming.

2.2.3. Effect Of Led Light Spectrum On Iot-Based Indoor Lettuce Farming

Jaya et al, conducted a study to explore the impact of various LED light spectra on the growth of *Lactuca sativa L.* in an indoor farming environment. The research highlights the importance of light quality in enhancing plant morphology and productivity. By examining the effects of white and yellow LED spectra, the study evaluates key growth parameters while monitoring environmental conditions through IoT technology. The experiment was carried out at the Computer Network Laboratory of Politeknik Negeri Lampung, using a hydroponic system with different varieties of lettuce. Several growth metrics, including leaf count, plant height, root length, and fresh weight at harvest, were measured and repeated for accuracy. The data were analyzed using a Student's t-test. The findings reveal that the yellow LED spectrum results in superior lettuce growth, with longer plants and improved overall development compared to the white LED spectrum. This research contributes to the understanding of how light quality influences plant growth in indoor farming, while also emphasizing the role of IoT technology in providing detailed monitoring and management. It aligns with existing studies that examine the relationship between light conditions and plant development.

2.2.4. Smart Soil Property Analysis Using IoT: A Case Study Implementation in Backyard Gardening

R. Aarthi et al, authored a paper titled "Smart Soil Property Analysis Using IoT: A Case Study Implementation in Backyard Gardening," which addresses the current challenges of soil fertility management in agriculture. The study focuses on issues related to fertilizer use in Indian farming, where many farmers apply fertilizers without proper soil analysis. The research aims to develop an IoT-based system designed to predict soil fertility and monitor key factors such as salinity, pH levels, temperature, and moisture in real time. The methodology involves a comprehensive system that integrates data collection, transmission, processing, and a smart application unit. This system was tested in the context of backyard gardening, specifically for growing spinach. The key findings highlight the prototype's ability to accurately monitor soil conditions, demonstrating minimal

discrepancies when compared to laboratory results. The authors conclude that the system developed in this study has the potential to improve soil management practices, enhance resource utilization, and boost agricultural productivity.

2.2.5. The management model of Ceking Terrace Tegallalang based on sustainable tourism development

Pande Putu Tobi et al, examined the management framework for the Ceking Terrace tourist destination in Tegallalang, Bali. The primary aim of this research was to evaluate management practices that enhance community well-being, promote environmental sustainability, and improve the tourist experience. A qualitative descriptive method was employed, including observations, documentation, and in-depth interviews with various stakeholders, such as local leaders and visitors. The findings revealed that the POAC (Planning, Organizing, Actuating, and Controlling) model plays a significant role in boosting village income. However, its impact on community development was found to be limited, with issues related to public facilities and employee performance negatively affecting the outcomes. The study also indicated that the management of Ceking Terrace successfully integrates sustainable tourism practices with economic, social, and environmental policies. However, it suggests that improvements in public facilities and better coordination among stakeholders are necessary to achieve a more substantial positive impact on both the community and tourists. This research contributes to the broader literature on sustainable tourism, underscoring the importance of community participation and environmental stewardship in tourism management.

2.2.6. Automatic Plant Watering System

Lokhande et al, developed an automated plant irrigation system designed to address the challenges of garden maintenance in institutions facing manpower shortages during the COVID-19 pandemic. The main objective of the system is to offer a user-friendly, efficient solution that automatically waters plants based on their specific moisture needs, thus conserving water and ensuring plant health. The system uses an Arduino UNO microcontroller and soil moisture sensors to monitor soil conditions, activating a motor to irrigate the plants when moisture levels drop. Key findings show that the system operates effectively only when the soil is dry, helping to prevent overwatering and reducing the consumption of water. Classified under smart irrigation, this research presents an

IoT-based solution tailored to the unique requirements of individual plants, unlike traditional irrigation methods. The article concludes that the Automatic Plant Watering System is a practical solution for urban dwellers and gardening enthusiasts, with potential for future enhancements, such as integrating additional sensors for soil health monitoring and a mobile app for user interaction. This underscores the system's relevance in the growing field of smart irrigation and IoT applications in agriculture.

2.2.7. Internet of Things and smart sensors in agriculture: Scopes and challenges

Prem Rajak et al, examines how the integration of IoT and smart sensors in agriculture can enhance crop yields and address various challenges faced by farmers. The study highlights the rationale for evaluating these technologies, particularly their effectiveness in monitoring environmental factors and improving agricultural practices. The methodology involves gathering data on key factors affecting crop yields using smart sensors and IoT devices. Key findings suggest that the implementation of IoT can enable more precise farming practices, improve resource management, and reduce the environmental impact. The study's contributions emphasize the importance of IoT and smart sensors in transforming agricultural practices, showcasing their potential to significantly boost productivity.

2.2.8. Ethnoecology and Climate Change Adaptation in Agriculture

Kamakaula et al, investigate the relationship between ethnoecology and agricultural adaptation techniques using a qualitative study approach, literature reviews, and thematic analysis. In order to find indigenous knowledge systems and traditional practices, the process entails examining academic publications, reports, and case studies. This method has the advantage of integrating several viewpoints and capturing rich, contextual insights. The preservation of ethnoecological knowledge in the face of socioeconomic pressures and cultural deterioration is a challenge, though. In order to create comprehensive adaptation methods and ultimately increase agricultural resistance to the effects of climate change, the study highlights the significance of fusing traditional knowledge with contemporary scientific research.

2.2.9. Design of an All-Purpose Terrace Farming Robot

Chapagain et al, The 'Aarohi' terrace farming robot, created by an IIT Kharagpur team, uses embedded technologies and sophisticated mechanical components to automate farming chores. Agronomic issues in smallholder agriculture, one of the key writers. To increase versatility, the approach entails creating separate, removable components for a range of farming tasks. Benefits include increased effectiveness in tackling issues like poverty and illiteracy in agriculture, as well as in negotiating rough terrain and carrying out a variety of duties. The intricacy of automation in limited settings and the requirement for reliable technology to guarantee dependability in various agricultural contexts are drawbacks, but Through the integration of cognitive systems, the robot seeks to transform terrace farming and increase output.

2.2.10. Globally Important Agricultural Heritage Systems in Japan: investigating selected agricultural practices and values for farmers

Nath et al, explores Japan's Globally Important Agricultural Heritage Systems (GIAHS). The authors used a mixed-methods approach, collecting information from key informants and farmers through surveys and structured questionnaires. Food and livelihood security, agro-biodiversity, local knowledge systems, cultural values, and landscape features were the five main criteria they used to evaluate the GIAHS values. The preservation of cultural heritage and improved ecological functioning are two benefits of GIAHS. None the less, difficulties were identified, including the requirement for local practice adaption and market forces. For sustainable growth, the study highlights how crucial it is to combine ancient knowledge with contemporary farming methods.

2.2.11. UAV Images for Agriculture Land Parcel Delineation through Edge Detection Algorithm: A Case Study of Hilly and Terai Regions

Bhomi et al, an edge detection technique was used to delineate agricultural land parcels using UAV photos. In order to maximise boundary extraction, the methodology tested several scale and merging levels. Post-processing methods such as boundary smoothing and breaking up smaller objects were then applied. This method is appropriate for normal agricultural management since it has the advantages of greater efficiency, less physical labour, and the capacity to outline parcels with a good degree of accuracy. But there are drawbacks, such as the requirement for cautious parameter selection and the possibility of errors in complicated terrains. All things considered, the study shows how UAV technology can improve land use planning and agricultural monitoring while emphasising how crucial algorithmic accuracy is to getting the best outcomes.

2.2.12. An AIoT-Based Automated Farming Irrigation System for Farmers in Limpopo Province

Relebogile et al, the study utilize an adapted Design Science Research Methodology (DSRM) to develop an AIoT-based Automated Farming Irrigation System. Key tools in the system include TensorFlow Lite for rainfall prediction and the Arduino ESP8266 microcontroller for data processing. The process follows six steps: problem identification, objective definition, design and development, demonstration, assessment, and communication. These steps ensure a structured approach to addressing the challenges of modern irrigation. The system offers several benefits, including timely irrigation based on rainfall forecasts, reduced over-irrigation, and more efficient water usage, all contributing to sustainable farming practices. By leveraging real-time data, the system enables farmers to make informed decisions, potentially improving crop yields and resource management. Additionally, the use of AI and IoT technologies allows for automation, minimizing human error and labor costs. However, potential drawbacks include reliance on technology and the need for farmers to adapt to new methods, which may require training and investment. Furthermore, there could be challenges in ensuring consistent connectivity in rural or remote areas. The overall aim of the research is to conserve water resources, enhance agricultural productivity, and contribute to the development of smarter, more sustainable farming systems. The study highlights the role of innovation in addressing water scarcity and improving the efficiency of agricultural practices.

2.2.13. Smart Farming for Efficient Crop Growth

Prathibha et al, examines the essential function of agriculture in India and the urgent need for contemporary methods that can boost productivity and sustainability through the incorporation of modern technologies, particularly the Internet of Things (IoT). The primary objective is to develop a smart farming system equipped with various sensors that operate on an Atmega 328 microcontroller to measure environmental factors such as soil moisture, temperature, and pH levels, enabling real-time data analysis to enhance crop management and identify diseases. A key finding of the study is that IoT significantly aids in monitoring crop health and environmental conditions. Crucial components include determining suitable crops based on soil pH and utilizing image processing for disease detection. This paper helps address a knowledge gap by offering a framework for the application of IoT technology in agriculture, where real-time data is vital for informed decision-making. Consequently, the authors affirm that smart farming practices help reconcile agricultural productivity with sustainability, making their findings particularly pertinent

to existing literature on precision agriculture and smart farming technologies.

2.2.14. European agricultural soil management: Towards climate-smart and sustainability, knowledge needs and research approaches

Keesstra et al, wrote the paper titled "European agricultural soil management: Towards climate-smart and sustainability." It covers the crucial importance of sustainable soil management in tackling environmental issues and fulfilling the goals established by the European Green Deal. The authors outline the study's aims and objectives, which include identifying knowledge gaps related to soil management practices and highlighting the necessity for research that may enhance the functions of soils in delivering these services. The research methodology involves a literature review, stakeholder consultations, and analysis of various reports connected to soil management on different scales within Europe. Major findings indicated significant knowledge and understanding gaps regarding the effects of soil management practices on ecosystem services, particularly emphasizing the need for cross-disciplinary collaboration and long-term studies. This paper enriches the literature by providing a roadmap for upcoming research in soil management, advocating for integrated soil management that considers socio-economic and environmental factors, while emphasizing the crucial need for collaboration among multiple stakeholders to formulate the most effective soil management strategies. By placing the findings within the broader context of existing literature on soil management and sustainability, the relevance to the literature survey is underscored, thus guiding future research and policy efforts aimed at achieving the goals of the European Green Deal.

2.2.15. Smart Crop Adaptability and Monitoring (SCAM)

Hazim Sakri et al, explores the use of IoT devices in agriculture, specifically focusing on an automated system for irrigation, cooling, and lighting that promotes plant growth while minimizing labor requirements. The study involved creating a model equipped with various sensors to monitor environmental factors. The collected data is processed through the Blynk application, allowing for real-time monitoring and management. Consequently, the main findings suggest that SCAM successfully automates agricultural tasks and adapts to variable conditions, despite encountering challenges related to sensor accuracy during testing. This research makes a valuable contribution to the area of smart agriculture by demonstrating the potential benefits of IoT technologies in enhancing efficiency and sustainability, thereby highlighting the importance of incorporating such innovations into contemporary farming methods.

The following table summarizes key research papers reviewed as a part of the literature survey. Each entry includes the title, author details, publication source and year of publication, along with remarks highlighting the relevance of each paper to our project objectives. This overview provides insights into existing studies, methodologies and identified gaps in the field.

TABLE 2.1 INFERENCES FROM LITERATURE SURVEY

S.No	Title of the paper	Author Details	Publication & Year	Remarks
1	Automatic Greenhouse Monitoring System	Shazeb,Mohammed Safeeruddin,Doijode Santosh, and Harpreet Bedi.	Indonesian Journal of Applied and Industrial Sciences, 2024	• The study demonstrates the potential of IoT in automating greenhouse monitoring for better resource efficiency and crop yield.
2	Sustainable Agriculture in the Digital Age: Crop Management and Yield Forecasting with IoT, Cloud, and AI	Ashraf, Hadeed, and Musliudeen Toheeb Akanbi	Tensor gate Journal of Sustainable Technology and Infrastructure for Developing Countries, 2023	• The paper highlights how IoT, Cloud, and AI are revolutionizing crop management and yield forecasting, driving sustainable agriculture.
3	Effect of Light Spectrum LED Lettuce on IoT-Based Indoor Farming.	Jaya, Moh Haris Imron S., Septafiansyah Dwi Putra, and Imam Sofi'i	Journal of Tropical Biology,2023	• The study demonstrates that different LED light spectra, particularly yellow LEDs, significantly enhance the growth of Lactuca sativa L. in indoor farming.
4	Smart soil property analysis using IoT: A case study	Aarthi, R., D. Sivakumar, and Vinayagam Mariappan	Procedia Computer Science,2023	• The study introduces an IoT system for real-time soil fertility analysis,

	implementation in backyard gardening			addressing fertilizer misuse in Indian farming.
5	The management model of Ceking Terrace Tegallalang based on sustainable tourism development	Sukmadiarta, Pande Putu Tobi, and Ni Nyoman Sunariani.	Review Of Management, Accounting, And Business Studies,2023	• The study emphasizes the importance of integrating economic, social, and environmental policies to enhance sustainable tourism at Ceking Terrace.
6	Automatic Plant Watering System	Lokhande, Sharayu A., Shilpa Pawar, and Neha Chauhan	Journal of Renewable Energy Exchange , 2023	• The study presents an efficient, automated irrigation system that conserves water and ensures plant health by meeting individual moisture needs.
7	Internet of Things and smart sensors in agriculture: Scopes and challenges	Rajak, Prem, Abhratanu Ganguly, Satadal Adhikary, and Suchandra Bhattacharya	Journal of Agriculture and Food Research,2023	• The paper explores how IoT and smart sensors can enhance crop yields by improving monitoring and precision farming techniques.
8	Ethnoecology and Climate Change Adaptation in Agriculture	Kamakaula, Yohanes	Global International Journal of Innovative Research,2024	• The paper examines how integrating indigenous knowledge with modern science can improve agricultural adaptation to climate change.

9	Design of an all-purpose terrace farming robot	Mohta, Vibhakar, Adarsh Patnaik, Shivam Kumar Panda, Siva Vignesh Krishnan	arxiv preprint arxiv, 2022	The 'Aarohi' robot integrates embedded technologies to automate farming tasks, improving efficiency in smallholder agriculture.
10	Globally Important Agricultural Heritage Systems in Japan: investigating selected agricultural practices and values for farmers	Nath, Tapan Kumar, Makoto Inoue, Yim Ee Wey, and Saori Takahashi.	International Journal of Agricultural Sustainability,2024	The paper assesses Japan's Globally Important Agricultural Heritage Systems (GIAHS), focusing on their importance to farmers and sustainability.
11	UAV Images for Agriculture Land Parcel Delineation through Edge Detection Algorithm: A Case Study of Hilly and Terai Regions	Bhomi, Arun Kumar, Jiya Thapa, Mamta Kadel, Nischal Acharya, Prawal Parajul	Journal on Geoinformatics, Nepal,2023	The study demonstrates how UAV-based edge detection can efficiently delineate agricultural land parcels with high accuracy.
12	An AIoT-Based Automated Farming Irrigation System for Farmers in Limpopo Province	Langa, Relebogile, Michael Nthabiseng Moeti, and Thabiso Maubane	Journal of Innovation Information Technology and Application,2024	The study utilizes an AIoT-based system with TensorFlow Lite and Arduino ESP8266 for efficient, data-driven irrigation.

13	Smart Farming for Efficient Crop Growth	Thirrunavukkarasu, R. R., M. Deepika, R. Dharshini, A. Infant Shiny, and S. Karthiga	ACS Journal for Science and Engineering,2023	• The study emphasizes the role of IoT in improving crop management by monitoring environmental factors like soil moisture, temperature, and pH levels.
14	European agricultural soil management: Towards climate-smart and sustainability, knowledge needs and research approaches	Keesstra, S. D., Claire Chenu, L. J. Munkholm, S. Cornu, P. J. Kuikman	European Journal of Soil Science,2024	• The paper emphasizes sustainable soil management as essential for addressing environmental challenges and meeting the European Green Deal goals.
15	Smart Crop Adaptability and Monitoring (SCAM)	Sakri, Muhammad Hazim Bin Mohd	Progress in Engineering Application and Technology,2023	• The study demonstrates the use of IoT devices to automate irrigation, cooling, and lighting, improving plant growth and reducing labor.

CHAPTER 3

SYSTEM REQUIREMENT

3.1 INTRODUCTION

The Terrace Farming System is designed to seamlessly integrate hardware and software components to monitor and manage urban terrace gardens effectively. The system uses IoT-based sensors to collect real-time environmental data and a mobile application to process and display this data in a user-friendly interface. The primary goal is to provide urban farmers with timely and accurate information that supports informed decision-making and efficient resource management.

The hardware includes sensors for measuring soil moisture, temperature, humidity, and light intensity, which are essential parameters for plant health and growth. These sensors are connected to a microcontroller (such as an Arduino or ESP32), which transmits the data to the cloud or directly to the mobile app via Wi-Fi or Bluetooth.

On the software side, the system features a mobile application built for the Android platform. It provides personalized recommendations, notifications, and visual dashboards based on real-time sensor readings. The application is designed to be intuitive and accessible, catering to users with varying levels of technical and agricultural knowledge.

Overall, the system specification emphasizes modularity, scalability and ease of use, ensuring that it can be deployed across different urban settings and adapted for various plant types and growing conditions.

3.2 HARDWARE REQUIREMENT

- Soil Moisture Sensor(YL-69)
- Temperature and Humidity Sensor(DHT22)
- Wi-Fi Module : Node MCU ESP8266
- Power Supply
- Voltage Regulator
- LCD Display

- Capacitor
- Resistor

3.3 SOFTWARE REQUIREMENT

- Android xml (for front-end)
- Java (for back-end)
- Blynk IoT App
- Firebase

3.4 SOFTWARE DESCRIPTION

Android XML

Android XML (Extensible Markup Language) is used to design the user interface (UI) of Android applications. It defines the layout and appearance of each screen using structured tags, allowing developers to visually organize components like buttons, text views, images, and input fields. In our Terrace Farming mobile application, Android XML is used to create a clean, responsive, and user-friendly front end. It helps display real-time sensor data (like temperature, humidity, and soil moisture) and provides an intuitive layout for features such as user registration, crop selection, and personalized farming guidance. By using XML for UI and Java for logic, the app maintains a clear separation between design and functionality.

Java

Java serves as the backend programming language for the Terrace Farming mobile application. It is responsible for handling the core logic and data processing behind the user interface. In this project, Java manages backend operations such as user authentication, communication with Firebase for real-time sensor data retrieval, and triggering alerts when environmental conditions cross specified thresholds. It also processes user input for features like profile setup and crop selection, enabling personalized farming

recommendations. By connecting the frontend (designed in Android XML) with the cloud database, Java ensures seamless data flow, secure operations, and reliable performance across the application.

Blynk IoT App

Blynk is an IoT platform that allows users to create custom mobile applications for controlling and monitoring hardware devices remotely. It provides a drag-and-drop interface where widgets such as buttons, gauges, graphs, and notifications can be easily configured to display real-time data. In our Terrace Farming project, the Blynk app is used to visualize sensor data from the microcontroller, including temperature, humidity, and soil moisture levels. It enables remote monitoring and control, allowing users to receive alerts and make decisions based on live environmental conditions. Blynk enhances the interactivity of the system by offering a reliable and user-friendly interface without the need for complex app development from scratch.

Firebase

Firebase is a cloud-based platform by Google that provides a suite of backend services for mobile and web applications. In the Terrace Farming project, Firebase is used to handle real-time database storage, user authentication, and cloud communication. Sensor data such as temperature, humidity, and soil moisture is sent from the microcontroller to Firebase, where it is stored and instantly synced with the mobile application. Firebase Authentication enables secure user login using email, phone number, or Google Sign-In. Its real-time capabilities allow users to monitor live farming conditions and receive instant alerts. Firebase ensures reliable, scalable, and secure backend support for the application.

3.5 HARDWARE DESCRIPTION

The Terrace Farming monitoring system is built using a combination of IoT hardware components that work together to collect, process, and transmit environmental data. The central unit of the system is a microcontroller, such as the ESP32 or Arduino Uno, which is responsible for interfacing with sensors and sending the collected data to the cloud or mobile application. To measure environmental parameters, the system uses a DHT22 sensor for capturing ambient temperature and humidity levels. These readings are crucial for maintaining the right conditions for plant growth. A soil moisture sensor is also integrated into

the system to monitor the water content in the soil. This helps in determining irrigation needs and avoiding water wastage. All sensors are connected to the microcontroller, which is powered using a regulated power supply or a rechargeable battery pack, ensuring consistent operation even in outdoor settings. For communication, the system uses the microcontroller's Wi-Fi module to connect with platforms like Firebase or Blynk, allowing real-time data transmission to the mobile app. The hardware is compact, lightweight, and suitable for deployment on rooftops and terraces. This configuration ensures efficient environmental monitoring and supports sustainable farming practices in urban areas.

CHAPTER 4

PROJECT DESCRIPTION

GAP IDENTIFIED FROM LITERATURE SURVEY

The literature review highlights several shortcomings in existing agricultural solutions. For instance, many integrated platforms do not effectively merge real-time monitoring, insights and guidance; in contrast, others do not adequately address the distinct challenges associated with space and resource limitations encountered by urban terrace farms. Currently available IoT systems often feature interfaces that are not user-friendly, lack tailored recommendations, or fail to provide a thorough integration of various resources, often emphasizing single crops or rural settings. Additionally, these systems do not adequately support scalability and community involvement, which hinders their broader adoption. There is a compelling need for the proposed study, as urban terrace farming requires specific systems that can tackle these unique issues and a cohesive IoT solution will offer real-time monitoring alongside personalized recommendations. This project aims to provide versatility for various crops and empower urban farmers with sustainable, data-driven solutions through a combined system of sensors and a mobile application centered on user experience. Focusing on optimizing resources to enhance productivity, the research intends to support urban agriculture in scalable and accessible manners, delivering solutions that can be readily embraced by individuals without technical expertise.

4.1 EXISTING SYSTEM

In urban areas, terrace farming is traditionally managed through manual methods, where individuals rely on physical observation to monitor soil moisture, temperature, and humidity. This manual approach often results in inefficient water usage, inconsistent plant care, and low crop yields, as it is difficult for users to accurately assess environmental changes in real time. While some mobile applications and smart farming technologies exist, they are mostly designed for large-scale agricultural operations and are not well-suited for small, rooftop-based farming. Furthermore, many existing systems are complex, expensive, and lack customization options for urban settings. They also often do not integrate real-time data from IoT sensors, limiting their ability to provide timely and precise recommendations. As a result, there is a clear gap in providing an accessible, cost-effective, and easy-to-use solution that caters specifically to the needs of terrace farmers, particularly in densely populated regions.

4.2 DRAW BACKS OF EXISTING SYSTEM

- **Manual Monitoring:** Traditional terrace farming methods rely heavily on manual observation, leading to human errors and delayed responses to plant needs.
- **Inefficient Resource Use:** Without real-time data, water and fertilizer usage are often poorly managed, causing wastage and affecting crop health.
- **Low Yield and Quality:** Inconsistent care due to lack of timely monitoring reduces both the quantity and quality of the harvest.
- **Complexity of Available Solutions:** Existing smart farming apps are often complicated, designed for large agricultural fields, and not user-friendly for small-scale urban farmers.
- **Lack of Real-Time Data:** Many current systems do not integrate IoT sensors for real-time environmental monitoring, making it harder to make quick, informed decisions.
- **High Cost:** Advanced farming systems are often expensive, making them inaccessible for individuals practicing terrace farming on a small scale.

4.3 PROPOSED METHOD

The proposed methodology combines a mixed-method approach, utilizing both quantitative data from IoT sensors and qualitative feedback from users to design a user-centric, integrated platform for urban terrace farming. The system will leverage IoT hardware (e.g., soil moisture and temperature sensors) for real-time data collection, with Python-based analytics and machine learning models to provide actionable insights and personalized recommendations, such as dynamic watering schedules. A mobile app, developed using frameworks like Flutter or React Native, will interface with the IoT system, offering a user-friendly experience for non-technical urban farmers. The project will integrate a decision support system (DSS) that combines sensor data, external environmental information, and machine learning algorithms to optimize resource use and improve productivity. This approach addresses gaps identified in the literature, such as the lack of integrated solutions and personalized guidance, with a focus on urban farming challenges and multi-crop adaptability. The agile development process will ensure the system evolves based on user feedback, ensuring practical usability.

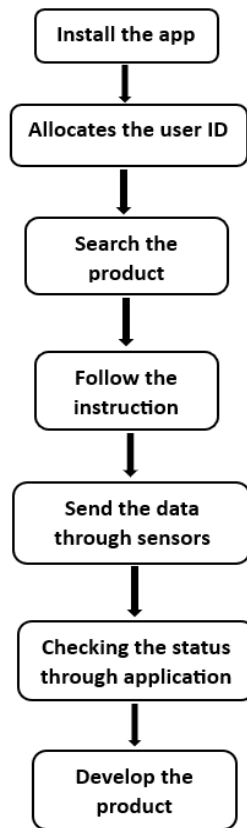


Fig 4.1 Consumer's Module

The Fig 4.1 tells about the product is guided and developed by installing the app, registering for a user ID, following step-by-step instructions to monitor water levels, control sensors via WiFi, and check status and weather conditions through mobile.

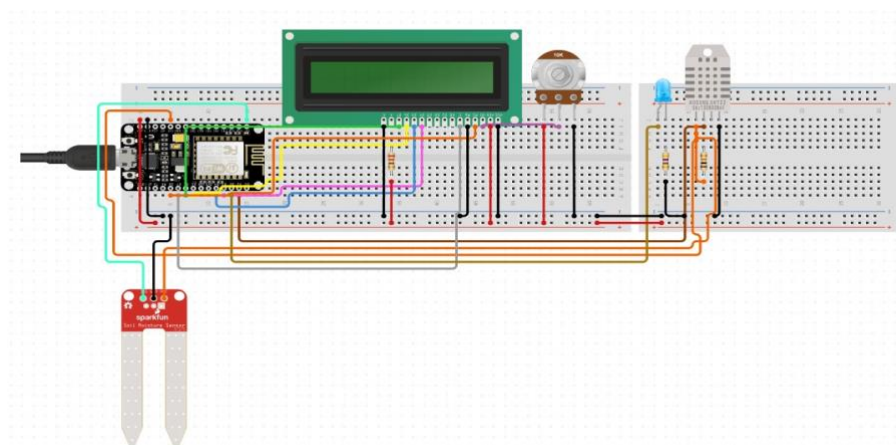


Fig 4.2 Circuit Diagram

The Fig 4.2 tells about this circuit connects an ESP8266 microcontroller with a soil moisture sensor, DHT11 temperature/humidity sensor, an LCD display, and an LED to monitor environmental data. The LCD shows real-time readings, while the LED can indicate moisture or temperature thresholds.

4.4 MODULE DESCRIPTION

4.4.1 User Registration and Profile Management Module:

This module allows users to create personalized accounts using secure authentication methods such as username and password. After successful registration, users can build detailed profiles by providing personal information, location details, and specific crop preferences. All profile data is securely stored and managed through Firebase, offering users a customized experience. Through this setup, users receive location-based farming tips, targeted recommendations, and timely agricultural reminders. This module ensures a seamless on boarding experience while maintaining high standards of data privacy and security.

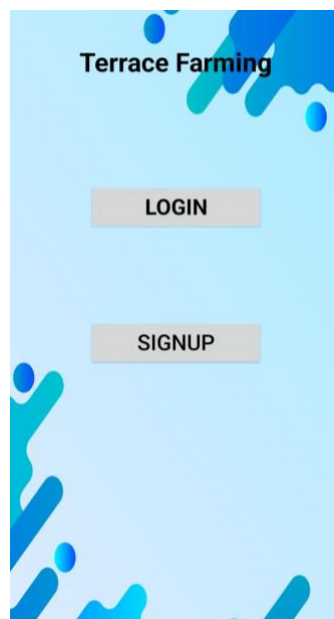


Fig 4.3 User Login Page

The Fig 4.3 tells about this is the login interface of a "Terrace Farming" Android app with two main options: LOGIN for existing users and SIGNUP for new users. It serves as the entry point to access farming data and features connected to the IoT-based system.

4.4.2 Real-Time Monitoring Module:

This module establishes a continuous link between IoT sensors (including soil moisture, temperature, and humidity sensors) and the mobile app via Node MCU ESP8266 microcontrollers. Real-time environmental data is collected, transmitted, and updated live in Firebase. Users can monitor critical factors like soil moisture content, ambient temperature, and air humidity directly from the app interface. This real-time tracking helps users make timely interventions to ensure optimal conditions for crop health and growth. The system is designed to operate efficiently with minimal latency, promoting proactive terrace farming practices.

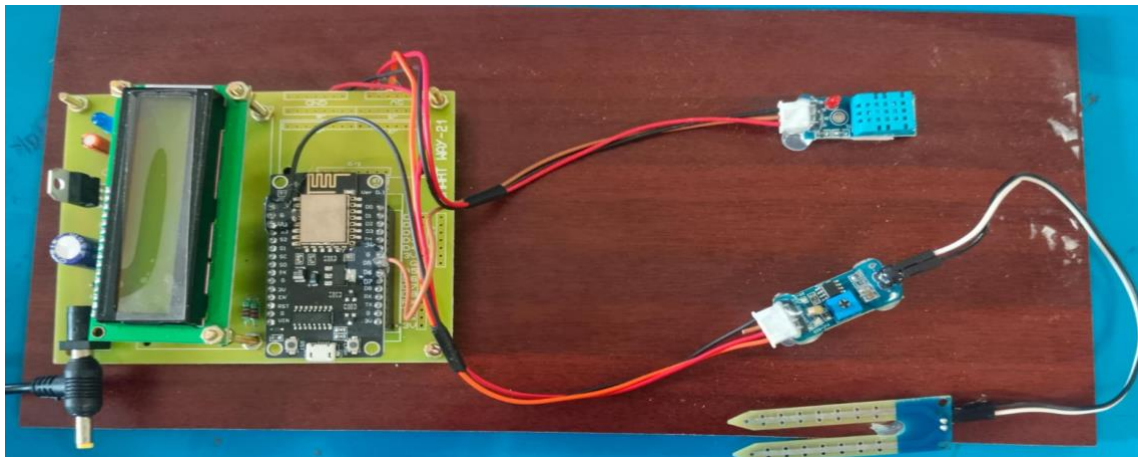


Fig 4.4 Real-Time Monitoring for Crop

The Fig 4.4 tells about this real-time monitoring setup uses an ESP8266 microcontroller with an LCD display, DHT11 sensor for temperature and humidity, and a soil moisture sensor. It continuously gathers environmental data and displays it on the LCD, making it ideal for terrace farming automation.

4.4.3 Data Analytics and Insights Module:

The Data Analytics module transforms raw sensor data into meaningful insights through powerful backend processing. It analyzes historical trends in moisture levels, temperature fluctuations, and user irrigation habits to uncover patterns. Users are provided with actionable insights like ideal watering schedules, early detection of unfavorable conditions, and suggestions to optimize crop yields. This module empowers farmers to transition from reactive to proactive decision-making, ensuring sustainable and efficient terrace farming operations.

Spinach
Tomato
Chilli
Okra
Carrot
Radish
Brinjal
Cucumber
Bell Pepper
Beetroot
Onion
Garlic
Lettuce

Fig 4.5 Data Analyzes

The Fig 4.5 tells about the system uses real-time sensor data (soil moisture, temperature, humidity) to monitor crop conditions via an Android app. Based on this data, it analyzes and suggests optimal crop choices and watering actions to improve terrace farming efficiency.

4.4.4 Water Management Module:

This module focuses on optimizing irrigation practices based on live soil moisture readings. Whenever the system detects low moisture levels, immediate notifications are sent to the user, helping prevent water stress for crops. Future upgrades to this module will include automated irrigation control by integrating solenoid valves and relay-based water systems, further minimizing human intervention. This smart management ensures that water resources are used efficiently, supporting environmentally friendly farming on terraces.

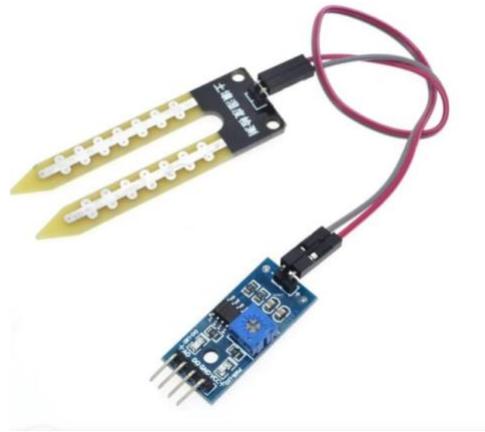


Fig 4.6 Moisture Sensor

The Fig 4.6 tells about this is a capacitive soil moisture sensor that measures the volumetric water content in soil. It sends analog or digital signals to a microcontroller (like NodeMCU) for real-time monitoring of soil moisture levels in smart farming systems.

4.4.5 Alerts and Notifications Module:

This module ensures users are always informed and ready to act. It delivers real-time alerts for critical events such as low soil moisture, extreme temperatures, and humidity issues. Additionally, it provides scheduled reminders for essential farming tasks like watering, fertilization, pruning, and harvesting. Notifications are personalized based on the user's profile and crop choices, ensuring that each user receives relevant and timely information. This system minimizes risks and maximizes productivity by promoting timely interventions.

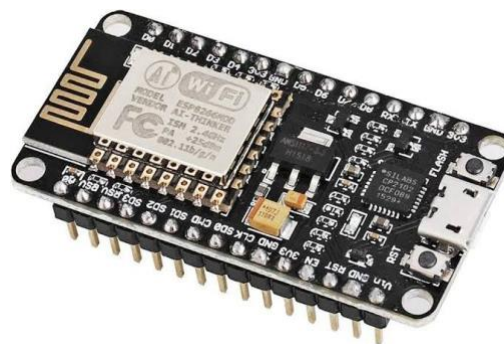


Fig 4.7 Node MCU

The Fig 4.7 tells about this is a NodeMCU ESP8266 Wi-Fi module used for IoT applications. It connects sensors like soil moisture and DHT11 to the internet for real-time monitoring and control via mobile or cloud platforms.

4.4.6 Integrated Device Management Module:

This module streamlines the setup, monitoring, and maintenance of all IoT devices involved in the system. It manages initial device configurations, maintains stable Wi-Fi and Firebase connectivity, and supports troubleshooting processes such as sensor re-calibration and network reconfiguration. It also allows users to easily add new devices or replace faulty sensors without technical complexity. This management ensures the system remains robust, scalable, and easy for users to maintain their terrace farming setup.



Fig 4.8 Monitoring in Blynk App

The Fig 4.8 tells about this Blynk app interface shows real-time monitoring of a terrace farming system, displaying sensor data like soil moisture, temperature, and humidity, along with a button to control irrigation remotely.

4.5 Advantages of Proposed Method

Real-Time Monitoring:

The system provides continuous real-time monitoring of temperature, humidity, and soil moisture using IoT sensors. This enables users to respond quickly to environmental changes, ensuring that plants

receive optimal care at the right time, which enhances overall plant health and supports better growth conditions throughout the farming cycle.

Efficient Resource Utilization:

By collecting and analyzing sensor data, the system helps manage water usage more efficiently. It ensures that irrigation occurs only when necessary, reducing water waste. This promotes sustainable farming practices and is especially beneficial in regions where water is scarce or expensive for small-scale urban farmers.

User-Friendly Mobile App:

The mobile application is designed with a simple and intuitive interface, allowing users of all skill levels to monitor farming conditions, receive alerts, and follow guided farming recommendations. Its accessibility makes it suitable for beginners and promotes the adoption of smart farming practices in urban environments.

Personalized Recommendations:

Based on the user's location and selected crop preferences, the system delivers customized farming advice. This helps users take specific actions suited to their region's climate and the requirements of individual crops, improving decision-making and maximizing productivity without needing advanced agricultural knowledge.

Low-Cost and Scalable:

The proposed system utilizes affordable hardware like open-source microcontrollers and low-cost sensors, making it budget-friendly for individuals and small communities. It is also easily scalable, allowing more sensors or features to be added over time without major redesigns, making it ideal for gradual expansion.

Reduced Manual Effort:

Automation significantly lowers the need for constant physical supervision. Farmers receive real-time alerts and insights directly through the mobile app, reducing time spent on manual inspections. This saves effort while ensuring plants are monitored effectively, even when users are away from their terrace farms.

Enhanced Crop Yield:

Accurate monitoring and timely interventions lead to improved plant health and growth. The system supports optimal conditions for each stage of plant development, resulting in better crop quality and higher yields. This directly benefits urban growers aiming for consistent and successful terrace farming outcomes.

CHAPTER 5

IMPLEMENTATION AND RESULT

5.1 OVERVIEW

Urban agriculture has gained momentum as cities face growing challenges related to food security, resource scarcity, and environmental sustainability. Terrace farming, a popular method in urban areas, allows individuals to cultivate crops in limited spaces such as rooftops and balconies. However, traditional terrace farming practices often rely on manual observation and generalized routines, which can lead to inefficient resource use and inconsistent yields.

To address these limitations, this project proposes a Terrace Farming System that integrates IoT sensors and a mobile application to enable real-time monitoring and personalized farm management. The system is designed to track critical environmental parameters such as soil moisture, temperature, humidity, and light intensity. Based on the collected data, the mobile app delivers data-driven recommendations to users, helping them optimize irrigation, schedule maintenance, and improve crop health.

The system promotes sustainable farming practices, enhances productivity, and simplifies terrace garden management, making it more accessible for urban dwellers—especially those with limited agricultural experience. This technology-driven approach not only reduces manual labor but also encourages efficient use of water and other resources, contributing to a smarter and greener urban lifestyle.

5.2 Model Performance

The Smart Terrace Farming System exhibited consistent and accurate performance during testing. The temperature, humidity, and soil moisture sensors effectively captured environmental data with reliable precision, ensuring timely detection of changes affecting plant health. The microcontroller (ESP32/Arduino) processed sensor inputs efficiently and maintained stable communication with the Firebase database and the mobile application. The mobile app displayed real-time updates with minimal latency, enabling users to monitor their terrace farm conditions instantly. The system promptly generated alerts when environmental parameters crossed critical thresholds, allowing users to take immediate corrective actions. Personalized crop recommendations were accurately delivered based on the user's profile and sensor data. Overall, the model proved to be a reliable, user-friendly, and cost-effective solution for urban terrace farming, with strong potential for real-world deployment and future scalability.

5.3 Project Output

The Terrace Farming system successfully monitors environmental conditions such as temperature, humidity, and soil moisture in real time using IoT sensors. The collected data is processed by a microcontroller and transmitted to a mobile application via Firebase. Users receive instant updates and alerts through the app, enabling timely actions like watering or crop adjustment. The system also provides personalized crop recommendations based on sensor readings and user preferences. Overall, the project delivers a reliable, user-friendly solution for urban farmers, promoting sustainable terrace farming by reducing manual effort, optimizing water usage, and improving crop care and yield.

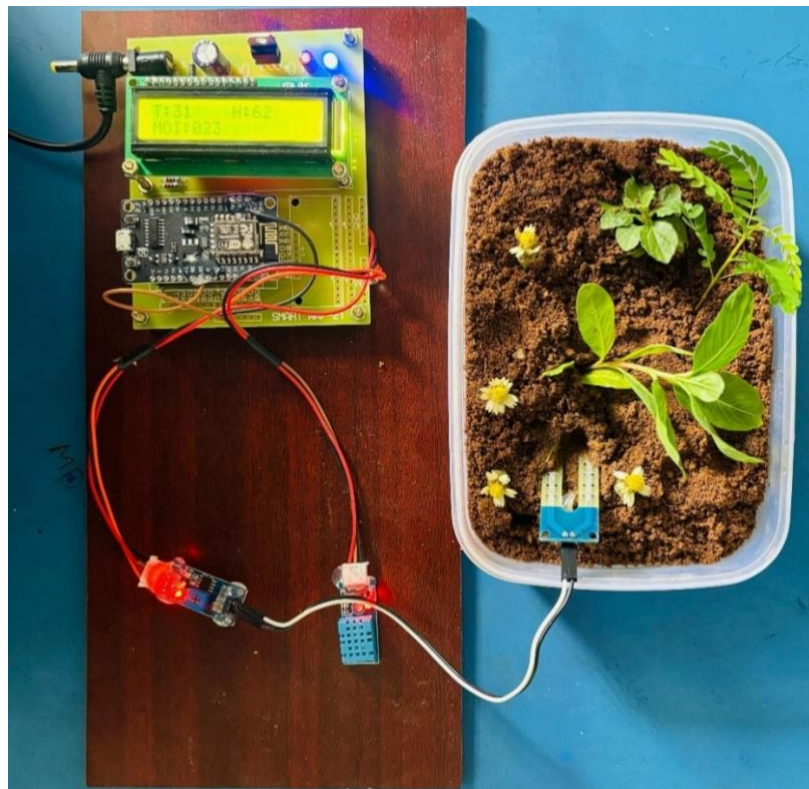


Fig 5.1 Crop Monitoring

The Fig 5.1 tells about this setup demonstrates a smart crop monitoring system using NodeMCU, a DHT11 sensor for temperature and humidity, and a soil moisture sensor, with live data displayed on an LCD and sent to the Blynk app.



Fig 5.2 Real-Time Data in Blynk App

The Fig 5.2 tells about this Blynk app screen shows real-time data from a terrace farming IoT system: temperature (31°C), humidity (64%), and soil moisture (47%), with an ON/OFF control button for automation.

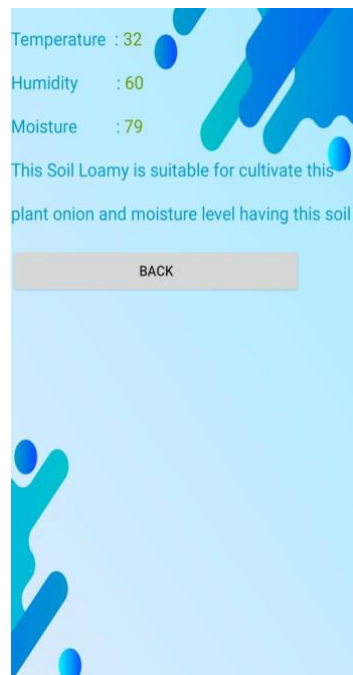


Fig 5.3 Output

The Fig 5.3 tells about this figure displays real-time sensor data (temperature, humidity, and moisture) collected and visualized in the Blynk App. Based on the sensor readings, the app provides a basic assessment of soil suitability for cultivation.

CHAPTER 6

CONCLUSION AND FUTURE WORK

CONCLUSION

The terrace farming system offers a transformative approach to urban agriculture. By leveraging IoT sensors and a user-friendly mobile platform, the system enables real-time monitoring of environmental conditions such as soil moisture, temperature, and humidity, ensuring optimal growth conditions for plants. The integration of data-driven insights helps urban farmers manage their crops effectively, regardless of their farming experience. The proposed system promotes resource efficiency, particularly in water usage, reducing waste and supporting sustainable farming practices. By providing personalized recommendations for watering and maintenance, the platform makes terrace farming more accessible and manageable for city dwellers, fostering self-reliant food production. Ultimately, this system not only contributes to improving urban food security but also aligns with environmental sustainability goals by reducing reliance on traditional farming resources. The combination of technology and traditional farming methods is a crucial step toward the future of urban agriculture.

FUTURE WORK

Future work on this smart terrace farming system involves several enhancements to increase its functionality and impact. First, integrating machine learning algorithms would allow the system to provide predictive insights, adapting to seasonal changes and user behavior, which would improve the accuracy of recommendations. Additionally, expanding the platform to support a broader range of plant species and varying environmental conditions would increase its usability for different types of terrace farms. Future versions of the system could also incorporate automated irrigation systems to complement the existing recommendations, reducing the need for manual intervention. To make the platform more accessible in areas with limited internet connectivity, offline capabilities could be integrated, ensuring continuous monitoring and guidance. Further improving user interaction with features like voice commands or integrating more intuitive user interfaces will enhance accessibility for a wider audience. Ultimately, the system could be scaled to support community gardens, vertical farming, and larger urban farming projects.

APPENDIX

SOURCE CODE

```
package com.prop.terracefarming;

import android.annotation.SuppressLint;
import android.app.ProgressDialog;
import android.content.Intent;
import android.os.AsyncTask;
import android.os.Bundle;
import android.util.Log;
import android.view.View;
import android.view.View.OnClickListener;
import android.widget.Button;
import android.widget.EditText;
import android.widget.Spinner;
import android.widget.Toast;


import java.io.BufferedReader;
import java.io.InputStreamReader;
import java.net.URL;
import java.net.URLConnection;
import java.sql.Connection;
import java.sql.DriverManager;
import java.sql.ResultSet;
import java.sql.SQLException;
import java.sql.Statement;


public class UserActivity extends MainActivity {

    Connection conn;

    EditText username,password,edtplant,edtcrop,edtplace,edtsize,edtdays;

    Button b1,b3,b2,bsensor;
```

```
String s1,s2,s3,pass1,plant,soil,crop,type,croptype,growth,weather,temperature;  
Spinner spinner1;
```

```
@Override
```

```
protected void onCreate(Bundle savedInstanceState) {  
    super.onCreate(savedInstanceState);  
    setContentView(R.layout.user_main);  
  
    b1=(Button)findViewById(R.id.btn1);  
    b2=(Button)findViewById(R.id.btn2);  
    b3=(Button)findViewById(R.id.btn3);  
  
    bsensor=(Button)findViewById(R.id.btnsensor);  
  
    edtcrop=(EditText)findViewById(R.id.edtcrop);
```

```
    spinner1 = (Spinner) findViewById(R.id.spinner1);  
    // conn=CONN();  
    b1.setOnClickListener(new OnClickListener(){
```

```
        @Override
```

```
        public void onClick(View v) {  
            // TODO Auto-generated method stub  
            crop=edtcrop.getText().toString();  
            new QuerySQL().execute();  
        }  
    }  
});
```

```
b2.setOnClickListener(new OnClickListener(){
```

```
        @Override
```

```

        public void onClick(View v) {
            // TODO Auto-generated method stub

            croptype= spinner1.getSelectedItem().toString();

            Intent intent=new Intent(UserActivity.this, CropResult.class);
            intent.putExtra("s1", croptype);
            startActivity(intent);
        }
    });

    b3.setOnClickListener(new OnClickListener(){

        @Override
        public void onClick(View v) {
            // TODO Auto-generated method stub

            startActivity(new Intent(UserActivity.this, UserLoginActivity.class));
        }
    });

    bsensor.setOnClickListener(new OnClickListener(){

        @Override
        public void onClick(View v) {
            // TODO Auto-generated method stub
            crop=edtcrop.getText().toString();
            new QuerySQL1().execute();
        }
    });
}

```

```

public class QuerySQL1 extends AsyncTask<String, Void, Boolean> {

    ProgressDialog pDialog ;
    Exception error;
    ResultSet rs;
    @Override
    protected void onPreExecute() {
        super.onPreExecute();

        pDialog = new ProgressDialog(UserActivity.this);
        pDialog.setTitle("Sensor");
        pDialog.setMessage("Retreiving values...");
        pDialog.setProgressStyle(ProgressDialog.STYLE_SPINNER);
        pDialog.setIndeterminate(false);
        pDialog.setCancelable(false);
        pDialog.show();
    }

    @Override
    protected Boolean doInBackground(String... args) {
        try {

            Class.forName("com.mysql.jdbc.Driver");
            conn = DriverManager.getConnection("jdbc:mysql://mysql-75344-
0.cloudclusters.net:18880/terracefarming","admin","cU5zYkH");

        } catch (SQLException se) {
            Log.e("ERRO1",se.getMessage());
        } catch (ClassNotFoundException e) {
            Log.e("ERRO2",e.getMessage());
        } catch (Exception e) {
            Log.e("ERRO3",e.getMessage());
        }
    }
}

```

```

        try {

            String COMANDOSQL="select * from cropdetails where
cropname='"+crop+"'";

            Statement statement = conn.createStatement();

            rs = statement.executeQuery(COMANDOSQL);
            if(rs.next()) {
                soil = rs.getString(4);
            }

            s1=
Read_URL("https://blynk.cloud/external/api/get?token=3115uKD3vmmFBX5DEDPqXjceXndVgL0s&
V0");

            s2=
Read_URL("https://blynk.cloud/external/api/get?token=3115uKD3vmmFBX5DEDPqXjceXndVgL0s&
V1");

            s3=
Read_URL("https://blynk.cloud/external/api/get?token=3115uKD3vmmFBX5DEDPqXjceXndVgL0s&
V2");

            return true;

            // Toast.makeText(getBaseContext(),
            // "Successfully Inserted.", Toast.LENGTH_LONG).show();
        } catch (Exception e) {
            error = e;
            return false;
        }

        // Toast.makeText(getBaseContext(),"Successfully Registered...",
        Toast.LENGTH_LONG).show();
    }

}

@SuppressLint("NewApi")
@Override
protected void onPostExecute(Boolean result1) {
    pDialog.dismiss ( ) ;
}

```

```

        if(result1)
        {

//                System.out.println("ELSE(JSON) LOOP EXE");

            try { // try3 open

                Intent intent=new Intent(UserActivity.this, UserActivity1.class);
                intent.putExtra("s1", s1);
                intent.putExtra("s2", s2);
                intent.putExtra("s3", s3);
                intent.putExtra("soil", soil);
                intent.putExtra("crop", crop);
                startActivity(intent);

            } catch (Exception e1) {
                Toast.makeText(getApplicationContext(), e1.toString(),
                    Toast.LENGTH_LONG).show();
            }

        }else
        {
            if(error!=null)
            {
                Toast.makeText(getApplicationContext(),error.getMessage().toString()
,Toast.LENGTH_LONG).show();
            }
            else
            {
                Toast.makeText(getApplicationContext(),"Check your credentials!!!"
,Toast.LENGTH_LONG).show();
            }
        }

        super.onPostExecute(result1);
    }
}

```

```

    }
}

private static String Read_URL(String theUrl) {
    StringBuilder content = new StringBuilder();
    try {
        URL url = new URL(theUrl);
        URLConnection urlConnection = url.openConnection();
        BufferedReader bufferedReader = new BufferedReader(new
InputStreamReader(urlConnection.getInputStream()));
        String line;
        while ((line = bufferedReader.readLine()) != null) {
            content.append(line + "\n");
        }
        bufferedReader.close();
    } catch(Exception e) {
        e.printStackTrace();
    }
    return content.toString();
}

public class QuerySQL extends AsyncTask<String, Void, Boolean> {
    ProgressDialog pDialog ;
    Exception error;
    ResultSet rs;

    @Override
    protected void onPreExecute() {
        super.onPreExecute();

        pDialog = new ProgressDialog(UserActivity.this);
        pDialog.setTitle("Searching");
        pDialog.setMessage("Search Crop Details...");
        pDialog.setProgressStyle(ProgressDialog.STYLE_SPINNER);
        pDialog.setIndeterminate(false);
    }
}

```

```

        pDialog.setCancelable(false);
        pDialog.show();
    }
    @Override
    protected Boolean doInBackground(String... args) {
        try {

            Class.forName("com.mysql.jdbc.Driver");

            conn = DriverManager.getConnection("jdbc:mysql://mysql-75344-
0.cloudclusters.net:18880/terracefarming","admin","cU5zYktH");

        } catch (SQLException se) {
            Log.e("ERRO1",se.getMessage());
        } catch (ClassNotFoundException e) {
            Log.e("ERRO2",e.getMessage());
        } catch (Exception e) {
            Log.e("ERRO3",e.getMessage());
        }
        try {

            String COMANDOSQL="select * from cropdetails where
cropname='"+crop+"'";

            Statement statement = conn.createStatement();
            rs = statement.executeQuery(COMANDOSQL);
            if(rs.next()){

                type = rs.getString(2);
                growth = rs.getString(3);
                soil = rs.getString(4);
                weather = rs.getString(5);
                temperature = rs.getString(6);
                return true;
            }
        }
    }

```

```

        return false;

        // Toast.makeText(getBaseContext(),
        // "Successfully Inserted.", Toast.LENGTH_LONG).show();
    } catch (Exception e) {
        error = e;
        return false;
    }
    // Toast.makeText(getBaseContext(), "Successfully Registered...",
    Toast.LENGTH_LONG).show();
}

}

@SuppressLint("NewApi")
@Override
protected void onPostExecute(Boolean result1) {
    pDialog.dismiss ( ) ;
    if(result1)
    {
        // System.out.println("ELSE(JSON) LOOP EXE");
        try { // try3 open

            Intent intent=new Intent(UserActivity.this,
SearchResultActivity.class);

            intent.putExtra("s1", type);
            intent.putExtra("s2", growth);
            intent.putExtra("s3", soil);
            intent.putExtra("s4", weather);
            intent.putExtra("s5", temperature);

            startActivity(intent);

        } catch (Exception e1) {
            Toast.makeText(getBaseContext(), e1.toString(),

```

```

Toast.LENGTH_LONG).show();

        }

    }else
    {
        if(error!=null)
        {
            Toast.makeText(getBaseContext(),error.getMessage().toString()
,Toast.LENGTH_LONG).show();
        }
        else
        {
            Toast.makeText(getBaseContext(),"No Matches Found!!!"
,Toast.LENGTH_LONG).show();
        }
    }
    super.onPostExecute(result1);
}
}

```

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